20285 Advanced Macroeconomics

Problem set 2

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Exercise 1: RBC Model

In this exercise we consider an RBC model. The basic set-up of the economy is as described below.

All markets are perfectly competitive, hence the representative household and the representative firm take as given the output price, normalised to 1 at each t, as well as input prices at each t.

Part 1.1: No-Ponzi Game Condition (NPGC)

We start with a version of the model where households can only save in the form of a risk-free one-period bond: each unit bought in period t pays a *certain* gross return R_{t+1} in $t+1$. Further, there is no uncertainty. Hence, the representative household problem is given by:

$$\max_{\{c_t, n_t, b_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \frac{n_t^{1+\phi}}{1+\phi} \right]$$

$$\text{s.t.} \begin{cases} c_t + b_{t+1} = w_t\, n_t + R_t\, b_t \\ b_0 \text{ given} \end{cases}$$

where c_t is consumption, n_t the amount of labour supplied by the household, w_t is the real wage, b_t is the amount of bonds held at the beginning of period t, and R_t is the return on bonds purchased at $t-1$, b_t. In this section, we consider a fixed gross interest rate on bonds, i.e., $R_t = R, \forall t \geq 0$.

We will specify the firm problem below, in Part 2.

1. Under the assumptions above, the flow budget constraint of the household is given by:

$$c_t + b_{t+1} = w_t\, n_t + R\, b_t.$$

By forward-iterating the flow budget constraint, derive the lifetime (intertemporal) budget constraint, i.e., an expression equating the present value (PV) of consumption to the initial wealth plus the PV of labour income, together with a limit term.

[HINT: The flow budget constraints at $t = 0$ and $t = 1$ are:

$$c_0 + b_1 = w_0 n_0 + R\, b_0$$
$$c_1 + b_2 = w_1 n_1 + R\, b_1$$

Combine the previous equations to get rid of b_1. Iterate on this procedure until $t = T$, take the limit $T \to \infty$. You should end up with an equation that contains three terms: (i) the PV of consumption, (ii) the PV of labour income, and (iii) a limit term related to the No-Ponzi game condition introduced below.]

Solution.

We derive the lifetime (intertemporal) budget constraint by induction.

Base case: The flow budget constraints at $t = 0$ and $t = 1$ are

$$c_0 + b_1 = w_0 n_0 + R b_0 \quad \text{and} \quad c_1 + b_2 = w_1 n_1 + R b_1$$
$$\iff$$
$$c_0 + b_1 = w_0 n_0 + R b_0 \quad \text{and} \quad b_1 = \frac{c_1 - w_1 n_1 + b_2}{R}$$
$$\iff$$
$$\frac{c_0 - w_0 n_0}{R} + \frac{c_1 - w_1 n_1}{R^2} + \frac{b_2}{R^2} = b_0.$$

Inductive hypothesis:

$$\sum_{t=0}^{T-1} \frac{c_t - w_t n_t}{R^{t+1}} + \frac{b_T}{R^T} = b_0$$

and

$$c_T + b_{T+1} = w_T n_T + R b_T \iff b_T = \frac{c_T - w_T n_T + b_{T+1}}{R}$$

implies

$$\sum_{t=0}^{T-1} \frac{c_t - w_t n_t}{R^{t+1}} + \frac{1}{R^T} \frac{c_T - w_T n_T + b_{T+1}}{R} = b_0$$
$$\iff$$
$$\sum_{t=0}^{T} \frac{c_t - w_t n_t}{R^{t+1}} + \frac{b_{T+1}}{R^{T+1}} = b_0$$
$$\iff$$
$$\sum_{t=0}^{T} \frac{c_t}{R^{t+1}} + \frac{b_{T+1}}{R^{T+1}} = b_0 + \sum_{t=0}^{T} \frac{w_t n_t}{R^{t+1}}$$

Therefore, by induction (and assuming $\sup_t c_t < \infty, \sup_t w_t < \infty$),

$$\sum_{t=0}^{T} \frac{c_t}{R^{t+1}} + \frac{b_{T+1}}{R^{T+1}} = b_0 + \sum_{t=0}^{T} \frac{w_t n_t}{R^{t+1}} \implies \sum_{t=0}^{\infty} \frac{c_t}{R^{t+1}} + \frac{b_{T+1}}{R^{T+1}} = b_0 + \sum_{t=0}^{\infty} \frac{w_t n_t}{R^{t+1}}.$$

We thereby derive an expression equating the present value of consumption to initial wealth plus the present value of labor income, together with a limit term:

$$\frac{1}{R} \sum_{t=0}^{\infty} \frac{c_t}{R^t} + \lim_{T\to\infty} \left(\frac{b_T}{R_T}\right) = b_0 + \frac{1}{R} \sum_{t=0}^{\infty} \frac{w_t n_t}{R^t}.$$

2. The No-Ponzi game (NPGC) imposes a restriction on the limit term derived above:

$$\lim_{T\to\infty} \left(\frac{1}{R}\right)^T b_T = 0.$$

Explain in words what this condition rules out, noting that $(1/R)^T b_T$ is the present value of bond holdings at date T. Why is such a condition needed in an infinite-horizon model with borrowing and lending? Finally, discuss why the household problem may fail to admit a *bounded* solution if the NPGC is not imposed.

[HINT: A sequence $\{x_t\}_{t=0}^{\infty}$ is bounded if $\exists\, M < \infty$ such that $|x_t| \leq M$ for all t. Start from a feasible candidate optimum, then increase c_0 slightly and adjust only $\{b_t\}$ to keep the flow constraints satisfied; without the NPGC, ask whether this deviation can remain feasible.]

Solution.

The No-Ponzi game condition imposes that the present value of bond holdings at date $T \to \infty$ is zero. In particular, the condition tells us that the agent cannot accumulate debt forever. Notice, however, that the condition does not require $\lim_{T\to\infty} b_T = 0$. Without the NPG, the problem does not admit a solution. The reason is that the agent will be able to borrow indefinitely.

To show it better, consider the problem without the NPG condition:

$$\max_{\{c_t, n_t, b_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \frac{n_t^{1+\phi}}{1+\phi}\right]$$

$$\text{s.t.} \begin{cases} c_t + b_{t+1} = w_t\, n_t + R_t\, b_t \\ b_0 \text{ given} \end{cases}$$

We show formally why such problem does not admit a solution. By contradiction, assume $(c_t^*, n_t^*, b_{t+1}^*)_{t=0}^{\infty}$ optimal. Now, consider $(c_t')_{t=0}^{\infty}$ such that $c_0' = c_0^* + \varepsilon$ with $\varepsilon > 0$ and $c_t' = c_t^*$ for all $t \geq 1$. Let $n_t' = n_t^*$ for all t. We want to show that there exists a sequence $(b_{t+1}')_{t=0}^{\infty}$ that satisfies all the 1-period resource constraints.

At $t = 0$, we had:

$$b_1^* = w_0 n_0^* + R b_0 - c_0^*$$

To maintain feasibility, if $c_0' = c_0^* + \varepsilon$, then $b_1' = b_1^* - \varepsilon$. At $t = 1$, feasibility requires:

$$\begin{aligned} b_2' &= w_1 n_1^* + R b_1' - c_1^* \\ &= w_1 n_1^* + R(b_1^* - \varepsilon) - c_1^* \end{aligned}$$

Hence, $b_2' = b_2^* - R\varepsilon$. Then, we can construct $b_{t+1}' = b_{t+1}^* - R^t\varepsilon$ such that $(c_t', n_t', b_{t+1}')_{t=0}^\infty$ is feasible and yield a strictly higher utility than $(c_t^*, n_t^*, b_{t+1}^*)_{t=0}^\infty$, reaching a contradiction. Note that, if the starting sequence $(c_t^*, n_t^*, b_{t+1}^*)_{t=0}^\infty$ happened to satisfy the NPG condition, the new sequence does not satisfy it:

$$\lim_{T\to\infty} \frac{1}{R^T} b_T' = \lim_{T\to\infty} \frac{1}{R^T}(b_T^* - R^{T-1}\varepsilon) = -\frac{\varepsilon}{R} \neq 0.$$

Of course, ε was arbitrary. A solution does not exist, much less a bounded solution, because the household is better off by increasing borrowing ε.

Part 1.2: Stochastic environment and capital

Next, we add two elements. First, we introduce uncertainty in the form of an exogenous productivity term in technology, z_t. Second, we add capital. The capital accumulation decision is made by the household, which implies the following updated maximization problem for the representative household:

$$\max_{\{c_t, n_t, b_{t+1}, k_{t+1}\}_{t=0}^\infty} \mathbb{E}_0 \sum_{t=0}^\infty \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \frac{n_t^{1+\phi}}{1+\phi}\right]$$

$$\text{s.t.} \begin{cases} c_t + b_{t+1} + k_{t+1} = w_t\, n_t + R_t\, b_t + R_t^k k_t + (1-\delta)k_t \\ k_0, b_0 \text{ given} \end{cases}$$

where k_t is the capital stock at the beginning of period t, δ is the capital depreciation rate, and R_t^k is the rental rate of capital.

The representative firm solves the following static problem:

$$\max_{y_t, n_t, k_t} \{y_t - w_t\, n_t - R_t^k k_t\}$$

$$\text{s.t.} \quad y_t = z_t\, n_t^{1-\alpha}\, k_t^\alpha$$

where y_t is the amount of output produced, n_t the amount of labour used in production, k_t the amount of capital used in production and z_t an exogenous productivity term.

3. Formulate the households maximization problem in sequential form, using the event-history notation covered in class. Specifically, let the exogenous productivity term depend on a generic stochastic process, i.e., $z_t(s^t)$. Hence, productivity at time t may depend on the *entire* history of stochastic events s^t.

   Solution.

   For simplicity, as in the lecture, assume that the state space is finite and that the family of probability measures $(\pi(s^t))_{t\geq 0}$ on the measurable space $(S^t, 2^{S^t})$ is Kolmogorov consistent $\forall t, \pi(s^t) = \sum_{s_{t+1}} \pi(s^t, s_{t+1})$. Then the Kolmogorov extension theorem guarantees that a probability measure $\mathbb{P}$ (and hence the expectation operator $\mathbb{E}_0$) over the measurable space $(S^{\mathbb{N}}, \mathcal{F})$ of infinite histories endowed with the sigma-algebra generated by cylinder sets is well-defined and consistent with $(\pi(s^t))_{t\geq 0}$ (see Billingsley 1986).

   Moreover, assume

$$\sum_{t=0}^\infty \beta^t \mathbb{E}_0 \left(\left| \frac{c_t(s^t)^{1-\sigma}}{1-\sigma} - \frac{n_t(s^t)^{1+\phi}}{1+\phi} \right| \right) < \infty.$$

   The assumption is a bit overkill, but it guarantees that

$$\mathbb{E}_0 \sum_{t=0}^\infty \beta^t \left(\frac{c_t(s^t)^{1-\sigma}}{1-\sigma} - \frac{n_t(s^t)^{1+\phi}}{1+\phi} \right) = \sum_{t=0}^\infty \beta^t \mathbb{E}_0 \left(\frac{c_t(s^t)^{1-\sigma}}{1-\sigma} - \frac{n_t(s^t)^{1+\phi}}{1+\phi} \right)$$

because by the triangle inequality

$$\left|\sum_{t=0}^{\infty}\beta^t\left(\frac{c_t(s^t)^{1-\sigma}}{1-\sigma}-\frac{n_t(s^t)^{1+\phi}}{1+\phi}\right)\right|\leq\sum_{t=0}^{\infty}\left|\beta^t\left(\frac{c_t(s^t)^{1-\sigma}}{1-\sigma}-\frac{n_t(s^t)^{1+\phi}}{1+\phi}\right)\right|\in L^1(S^{\mathbb{N}},\mathcal{F},\mathbb{P})$$

and hence by dominated convergence

$$\begin{aligned}&\mathbb{E}_0\sum_{t=0}^{\infty}\beta^t\left(\frac{c_t(s^t)^{1-\sigma}}{1-\sigma}-\frac{n_t(s^t)^{1+\phi}}{1+\phi}\right)\\&=\sum_{t=0}^{\infty}\mathbb{E}_0\beta^t\left(\frac{c_t(s^t)^{1-\sigma}}{1-\sigma}-\frac{n_t(s^t)^{1+\phi}}{1+\phi}\right)\\&=\sum_{t=0}^{\infty}\sum_{s^t\in S^t}\beta^t\pi(s^t)\left[\frac{c_t(s^t)^{1-\sigma}}{1-\sigma}-\frac{n_t(s^t)^{1+\phi}}{1+\phi}\right].\end{aligned}$$

Therefore, assuming that $c_t, n_t, b_{t+1}, k_{t+1}$ are 2^{S^t} measurable (u is trivially $\mathcal{B}(\mathbb{R}^2)$ measurable because continuous), we can write

$$\max_{(c_t(s^t),n_t(s^t),b_{t+1}(s^t),k_{t+1}(s^t))_{t=0,s^t\in S^t}^{\infty}}\sum_{t=0}^{\infty}\sum_{s^t\in S^t}\beta^t\pi(s^t)\left[\frac{c_t(s^t)^{1-\sigma}}{1-\sigma}-\frac{n_t(s^t)^{1+\phi}}{1+\phi}\right]$$

$$\text{s.t.}\forall t\geq 0, s^t\in S^t\begin{cases}c_t(s^t)+b_{t+1}(s^t)+k_{t+1}(s^t)=\\ \qquad\qquad w_t(s^t)n_t(s^t)+R_t(s^t)b_t(s^{t-1})+R_t^k(s^t)k_t(s^{t-1})+(1-\delta)k_t(s^{t-1})\\ k_0,b_0\text{ given}\end{cases}$$

4. Now, consider the household problem in recursive form. Does this require imposing more structure on the stochastic nature of the exogenous productivity term z_t? Provide an example of a stochastic process for z_t that allows for the household problem to be written in recursive form. How can the event-history notation be modified to reflect this new structure?

Solution.

In order to incorporate exogenous stochastic shocks into dynamic problems and preserve the recursive structure, we must require that this stochastic process has a recursive structure as well. Shocks that have this structure are called Markov processes. More precisely, we start with some definitions (see Stokey, Lucas, and Prescott 1989).

Definition 1. Let $(Z,\mathcal{Z})$ be a measurable space. A *transition function* (also called a Markov kernel) is a function $Q: Z\times\mathcal{Z}\to[0,1]$ such that:

(a) for each $z\in Z$, $Q(z,\cdot)$ is a probability measure on $(Z,\mathcal{Z})$; and
(b) for each $A\in\mathcal{Z}$, $Q(\cdot,A)$ is a $\mathcal{Z}$-measurable function.

The interpretation is that $Q(z,A)=\Pr\{z'\in A\mid z_t=z\}$: given that the current shock is z, the probability that next period's shock falls in the set A is $Q(z,A)$.

Definition 2. A stochastic process $\{z_t\}_{t=0}^{\infty}$ on $(Z,\mathcal{Z})$ is a *(first-order) Markov process* with transition function Q if, for all $t\geq 0$ and all $A\in\mathcal{Z}$,

$$\Pr(z_{t+1}\in A\mid z_0,z_1,\ldots,z_t)=\Pr(z_{t+1}\in A\mid z_t)=Q(z_t,A).$$

That is, the conditional distribution of next period's shock depends on the history only through the current value z_t. The process is called *stationary* (or *time-homogeneous*) if the same transition function Q governs transitions in every period.

Together, the Markov and stationarity properties ensure that the current state is a sufficient statistic for all future outcomes. This is precisely what makes the Bellman equation time-invariant and the optimal policy rule stationary (see Stokey, Lucas, and Prescott 1989).

Two special cases are worth noting:

- **i.i.d. shocks:** $Q(z, A) = \lambda(A)$ for all $z \in Z$, i.e., the transition function does not depend on the current state. This is a degenerate Markov process.
- **Higher-order Markov processes:** If the distribution of z_{t+1} depends on $k > 1$ lags, the state space can always be expanded by including those lags as additional state variables, reducing the system to a first-order Markov process.

The assumptions required specifically *on the stochastic process* for a recursive formulation to be valid are therefore the following:

(a) The shocks $\{z_t\}$ follow a **stationary, first-order Markov process** with transition function Q, as in Definitions 1 and 2.

(b) $Q(\cdot, A)$ is **measurable** for each $A \in \mathcal{Z}$ (condition 2 of Definition 1), which ensures that the integral in the Bellman equation is well defined.

The easiest case consists in assuming that productivity z_t follows a Markov chain process. In particular: 1) z_t does not depend on the full past history, but only on the current state, and 2) transition probabilities are invariant. Hence:

$$\pi(s_{t+1}|s^t) = \pi(s_{t+1}|s_t, s_{t-1}, \dots) = \pi(s_{t+1}|s_t).$$

5. Given the stochastic process for z_t provided by you in the previous point, write the households problem in recursive form and derive the associated optimality conditions. Clearly identify the exogenous and endogenous state variables, and the control variables in the problem. Use the envelope conditions to derive the Euler equations, and comment on all the optimality conditions you obtain.

Solution. The exogenous state variables are the aggregate bond holdings B, the aggregate capital stock K, and the aggregate labor supply N, the present state s, and the technology level z. The endogenous state variables are the household's bond holdings b and capital stock k. The control variables are next period bond holdings b', present consumption c, next period capital stock k', and present labor supply n.

The household's problem in recursive form is

$$V(b, B, k, K, N, s, z) = \max_{b', c, k', n} \left\{ \frac{c^{1-\sigma}}{1-\sigma} - \frac{n^{1+\phi}}{1+\phi} + \beta \sum_{s'} \pi(s'|s) V(b', B', k', K', N', s', z') \right\}$$

$$\text{s.t.} \begin{cases} c + b' + k' = nw(K, N) + R(B)b + R^k(K, N)k + (1-\delta)k \\ c, k' \geq 0 \\ 1 \geq n \geq 0 \\ B' = F(B) \\ K' = G(K) \\ N' = H(N). \end{cases}$$

The Lagrangian is

$$\mathcal{L} = \frac{c^{1-\sigma}}{1-\sigma} - \frac{n^{1+\phi}}{1+\phi} + \beta \sum_{s'} \pi(s'|s) V(b', B', k', N', s', z')$$
$$+\lambda\Big(w(K,N) + R(B)b + R^k(K,N)k + (1-\delta)k - k' - b' - c\Big)$$

and the FOCs are

$$\frac{\partial \mathcal{L}}{\partial c} = 0 \iff c^{-\sigma} = \lambda$$
$$\frac{\partial \mathcal{L}}{\partial n} = 0 \iff n^{\phi} = w\lambda$$
$$\frac{\partial \mathcal{L}}{\partial b'} = 0 \iff \beta \sum_{s'} \pi(s'|s) \frac{\partial V(b', B', k', K', N', s', z')}{\partial b'} = \lambda$$
$$\frac{\partial \mathcal{L}}{\partial k'} = 0 \iff \beta \sum_{s'} \pi(s'|s) \frac{\partial V(b', B', k', K', N', s', z')}{\partial k'} = \lambda.$$

We note here that in general, in order to take the FOC under the expectation operator, we implicitly assume that

(a) The value function is integrable for every k', b'

(b) The value function is differentiable at the optimal k', b', π *a.e.*

(c) There exists a nonnegative integrable function g such that in a neighbourhood of the optimal b', k', the absolute value of the partial derivative of the value function with respect to b' or k' is dominated by g, π *a.e.* (see Gall 2022).

This is all immediately satisfied in the finite case given our standard assumptions.

The relevant envelope conditions are

$$\frac{\partial V}{\partial b} = \lambda R$$
$$\frac{\partial V}{\partial k} = \lambda(R^k + (1-\delta)).$$

Combining the envelope conditions and the FOCs, we have

$$n^{\phi} = wc^{-\sigma} \tag{1.1}$$
$$\beta \mathbb{E}_s(R' c'^{-\sigma}) = c^{-\sigma} \tag{1.2}$$
$$\beta \mathbb{E}_s((R'^k + 1 - \delta)c'^{-\sigma}) = c^{-\sigma}. \tag{1.3}$$

Equation (1.1) is the intratemporal labor supply condition. It states that at the optimum the marginal disutility of labor is equal to the of the marginal benefit of labor, which is the wage of labor times the marginal utility of consumption. It represents the intratemporal tradeoff of leisure versus consumption.

Equation (1.2) is the intertemporal Euler equation for bonds. It states that at the optimum the marginal utility of consumption today is equal to the discounted expected marginal utility tomorrow times the gross return on bonds, where R is the gross return on investing (the intertemporal price of consumption). It represents the intertemporal tradeoff of consumption today versus consumption tomorrow. We can also rewrite that condition as follows:

$$1 = \mathbb{E}_t\left[R_{t+1}\beta \frac{u'(c_{t+1})}{u'(c_t)}\right] = \mathbb{E}_t\left[R_{t+1}Q_{t,t+1}\right]$$

where $Q_{t,t+1}$ is the Stochastic Discount Factor.

Equation (3) is the intertemporal Euler equation for capital. It states that at the optimum the marginal utility of consumption today is equal to the discounted expected marginal utility tomorrow times the gross return on capital, where R^k is the gross return on saving (the intertemporal price of consumption) and $1-\delta$ is the depreciation rate. It also represents the intertemporal tradeoff of consumption today versus consumption tomorrow.

6. Argue that $b_t = 0, \forall t \geq 0$, in the stochastic economy. Carefully explain the role that R_t plays for this result to hold. Does this condition ($b_t = 0, \forall t$) also hold in the deterministic economy discussed in part 1, where there is no capital?

   [HINT: Recall that bond supply must equal bond demand, and that all households make the same decisions since they are identical, or equivalently, there is a single representative household.]

   Solution.

   In the equilibrium, the rental rate for capital is positive (the production function has positive marginal products, so if the rental rate weren't positive, then factor demands would become unbounded, see Krueger 2012). Therefore, the intertemporal price of consumption is positive. The optimality conditions then imply that the interest rate on bonds would be positive (both the rental rate for capital and the interest rate on bonds represent the intertemporal price of consumption). But if the interest rate on bonds is positive, the representative household demands bonds. Since bond supply must equal bond demand, and all households make the same decisions since they are identical, nobody supplies bonds. Therefore, in the stochastic economy, $\forall t \in \mathbb{N}_0, b_t = 0$.

   Alternatively, and more concisely, since bond market clears $\forall t, B_t = 0$ and since there is a single representative household, it follows $\forall t, b_t = B_t = 0$.

   This condition also holds in the deterministic economy discussed in part 1, for the same reason: Either all households demand bonds, and nobody supplies, or all households supply bonds, and nobody demands, so that $\forall t \in \mathbb{N}_0, b_t = 0$.

7. Derive the optimality condition for labour and capital demand of the representative firm, and provide the economic intuition.

   Solution.

   The representative firm solves the static optimization problem $\forall t \in \mathbb{N}_0$

$$\max_{y_t, n_t, k_t} \{y_t - w_t n_t - R_t^k k_t\} \quad \text{s.t. } y_t = z_t n_t^{1-\alpha} k_t^{\alpha}$$

   that is

$$\max_{n_t, k_t} \{z_t n_t^{1-\alpha} k_t^{\alpha} - w_t n_t - R_t^k k_t\}.$$

   The FOCs are $\forall t \in \mathbb{N}_0$

$$z_t (1-\alpha) n_t^{-\alpha} = w_t \tag{1.4}$$

$$z_t \alpha k_t^{\alpha-1} = R_t^k. \tag{1.5}$$

   Equation (1.5) states that the price for renting capital is equal to the marginal product of capital. Equation (1.4) states that the price for buying labor is equal to the marginal product of labor.

Exercise 2: Preferences, Curvature, and labour Supply Elasticities

This exercise studies how the curvature of preferences over consumption shapes macroeconomic behavior. In Part A, we focus on the Constant Relative Risk Aversion (CRRA) specification for the consumption component of utility and show how its curvature parameter governs both risk attitudes and intertemporal substitution. In Part B, we turn to labour supply and examine how wage changes affect hours worked through substitution and income effects. Together, the two parts illustrate how preference curvature influences both consumption dynamics and labour supply elasticities in standard macroeconomic models.

Consider a representative household with preferences

$$\sum_{t=0}^{\infty} \beta^t \, u(c_t, n_t), \qquad \beta \in (0,1),$$

where the period utility function satisfies $u_c > 0$, $u_{cc} < 0$,

$$\lim_{c \to 0} u_c = \infty, \qquad \lim_{c \to \infty} u_c = 0,$$

as well as $u_n < 0$, $u_{nn} < 0$, with

$$\lim_{n \to 0} u_n = 0, \qquad \lim_{n \to \infty} u_n = -\infty.$$

In particular, we assume the following functional form:

$$u(c_t, n_t) = \frac{c_t^{1-\sigma} - 1}{1-\sigma} - \frac{n_t^{1+\phi}}{1+\phi}, \qquad \sigma \geq 0, \qquad \phi \geq 0.$$

As shown in PS1, in the limit as $\sigma \to 1$, utility converges to

$$u(c_t, n_t) = \log c_t - \frac{n_t^{1+\phi}}{1+\phi}.$$

Part 2.1: Constant Relative Risk Aversion (CRRA) Utility

In this section, we focus on the properties of the utility function with respect to consumption.

1. The coefficient of Relative Risk Aversion (RRA) is defined as

$$RRA(c) \equiv -\frac{c \, u_{cc}(c)}{u_c(c)}.$$

This object summarizes the household's attitude toward risk: a higher value of $RRA(c)$ corresponds to greater aversion to proportional (percentage) risk.

Consider a household with certain consumption $\bar{c}$. Suppose instead the household faces a small *proportional* consumption gamble that pays

$$\bar{c}(1+\varepsilon) \quad \text{or} \quad \bar{c}(1-\varepsilon)$$

with equal probability, where $\varepsilon > 0$ is small.

The (relative) *risk premium* π is defined as the fraction of $\bar{c}$ that the household would be willing to give up, accepting $\bar{c}(1-\pi)$ for sure, in order to avoid the gamble.

For small ε, the following approximation holds:

$$\pi \approx \frac{1}{2} RRA(\bar{c})\, \varepsilon^2.$$

You may take this approximation as given.

Compute $u_c(c)$ and $u_{cc}(c)$ for the CRRA utility function and show that

$$RRA(c) = \sigma \qquad \text{for all } c > 0.$$

Using the approximation above, explain in words what it means that $RRA(c)$ is constant under CRRA preferences. In particular, what does $RRA(c) = \sigma$ imply about how the willingness to pay to avoid a fixed percentage consumption gamble depends on the level of consumption $\bar{c}$?

Solution.

Let

$$u(c, n) = \frac{c_t^{1-\sigma} - 1}{1 - \sigma} - \frac{n_t^{1+\phi}}{1 + \phi}$$

for $\sigma > 0, \phi > 0$.

Then

$$u_c(c, n) = c^{-\sigma}$$

and

$$u_{cc}(c, n) = -\sigma c^{-\sigma-1}$$

and therefore

$$RRA(c) = \frac{-c u_{cc}(c)}{u_c(c)} = \frac{c\sigma c^{-\sigma-1}}{c^{-\sigma}} = \sigma.$$

Using the approximation above:

$$\pi \approx \frac{1}{2} RRA(\bar{c})\varepsilon^2 = \frac{1}{2}\sigma\varepsilon^2.$$

The coefficient of relative risk aversion $RRA(c)$ constant under CRRA preferences means that household attitudes toward risk do not depend on consumption levels. The willingness to pay to avoid a fixed percentage consumption gamble is independent of the level of consumption.

This is a strong behavioral assumption. Intuitively, a person with ten million dollars is (much) less willing to pay a fixed cost to avoid a gamble to win or lose one dollar than a person with ten dollars.

2. Define lifetime utility as

$$U(c, n) = \sum_{t=0}^{\infty} \beta^t u(c_t, n_t).$$

The marginal rate of substitution between consumption at t and $t + 1$ is

$$\text{MRS}_{t,t+1} = \frac{\partial U / \partial c_{t+1}}{\partial U / \partial c_t}.$$

The intertemporal elasticity of substitution (IES) is defined as

$$\text{IES} \equiv -\left(\frac{d\ln \text{MRS}_{t,t+1}}{d\ln\left(\frac{c_{t+1}}{c_t}\right)}\right)^{-1}.$$

In words, the IES measures the willingness to reallocate consumption across time: it captures how responsive the consumption growth rate c_{t+1}/c_t is to changes in the marginal rate of substitution between t and $t+1$.

Show that under CRRA utility the IES is

$$\text{IES} = \frac{1}{\sigma}.$$

Briefly interpret the result: what does a larger value of σ imply about (i) the curvature of $u(c)$ and (ii) the households willingness to substitute consumption across time?

Solution.

Recall that

$$u(c,n) = \frac{c_t^{1-\sigma}-1}{1-\sigma} - \frac{n_t^{1+\phi}}{1+\phi}$$

for $\sigma > 0, \phi > 0$.

Then

$$\text{MRS}_{t,t+1} = \frac{\beta^{t+1}c_{t+1}^{-\sigma}}{\beta^t c_t^{-\sigma}}$$

$$\frac{d\ln \text{MRS}_{t,t+1}}{d\ln(\frac{c_{t+1}}{c_t})} = (\frac{c_{t+1}}{c_t})\frac{d}{d(\frac{c_{t+1}}{c_t})}(-\sigma)\ln(\beta\frac{c_{t+1}}{c_t}) = (\frac{c_{t+1}}{c_t})(-\sigma)\frac{\beta}{\beta\frac{c_{t+1}}{c_t}} = -\sigma$$

and therefore

$$\text{IES} = -(-\sigma)^{-1} = \frac{1}{\sigma}.$$

A larger value of σ implies that the within-period utility function $u(c)$ is flatter (see Figure 2-1). More within-period consumption does not make the representative household (much) better off. Hence a larger value of σ implies that households are less willing to substitute consumption across time (see Figure 2-2). As σ increases from 0 to ∞, preferences shift from linear to Leontief. Within-period consumption shifts from perfectly substitutable to perfectly complementary.

3. Take as given the Euler equation

$$\frac{u_c(c_{t+1},n_{t+1})}{u_c(c_t,n_t)} = \frac{1}{\beta R_{t+1}}.$$

Using the definition of $\text{MRS}_{t,t+1}$ above, show that the Euler equation implies

$$\text{MRS}_{t,t+1} = \frac{1}{R_{t+1}}.$$

Finally, explain how the IES governs the responsiveness of the consumption growth rate c_{t+1}/c_t to changes in the intertemporal price $1/R_{t+1}$. In particular, how does a larger IES affect the sensitivity of consumption growth to movements in interest rates?

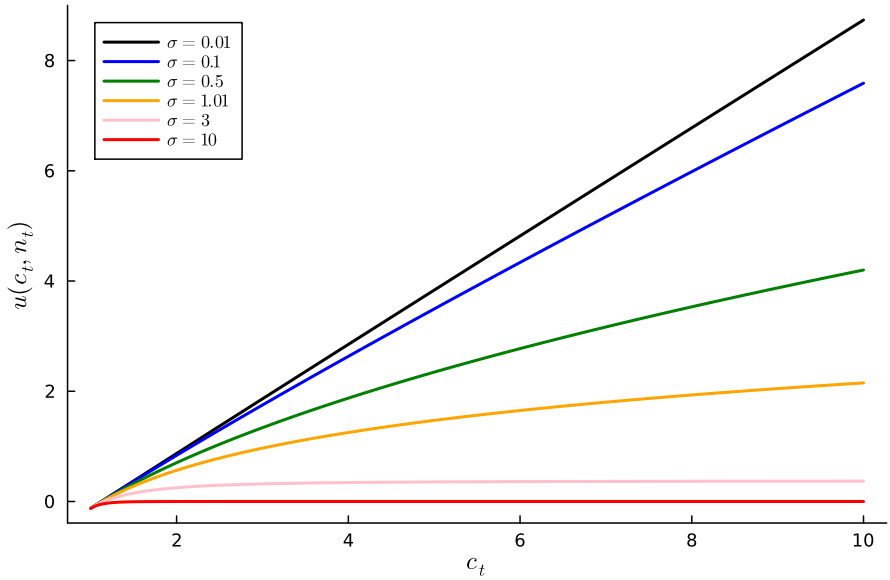


Figure 2-1. A larger value of σ implies that $u(c)$ is flatter

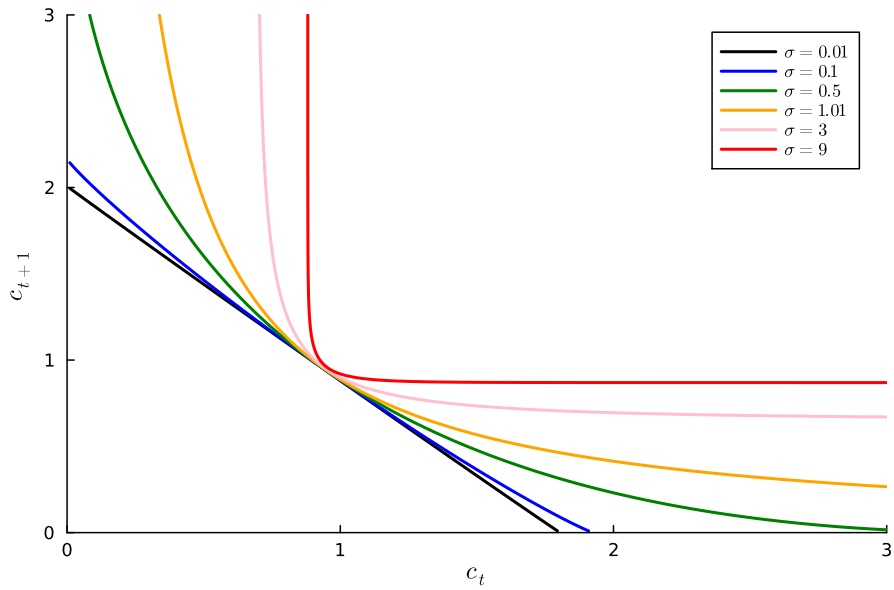


Figure 2-2. A larger value of σ implies that households are less willing to substitute consumption across time

Solution.

Recall that

$$\text{MRS}_{t,t+1} = \frac{\beta^{t+1} c_{t+1}^{-\sigma}}{\beta^t c_t^{-\sigma}} = \beta \frac{c_{t+1}^{-\sigma}}{c_t^{-\sigma}} = \beta \frac{u_c(c_{t+1}, n_{t+1})}{u_c(c_t, n_t)}.$$

Therefore

$$\text{MRS}_{t,t+1} = \beta \frac{u_c(c_{t+1}, n_{t+1})}{u_c(c_t, n_t)} = \frac{\beta}{\beta R_{t+1}} = \frac{1}{R_{t+1}}.$$

By this equality, if the gross interest rate R_{t+1} increases, then the marginal substitution rate $\text{MRS}_{t,t+1}$ decreases. If σ is large, then IES is small, and hence the consumption growth rate is not very responsive to changes in the intertemporal price $1/R_{t+1}$. If σ is small, then IES is large, and hence the consumption growth rate is very responsive to changes in the intertemporal price $1/R_{t+1}$.

A larger IES (a smaller σ) means consumption growth is more sensitive to movements in interest rates, and therefore consumers are more willing to substitute consumption between periods to take advantage of better interest rates.

From a different perspective, the lower the elasticity is, the larger is the change in in the interest rate needed to convince the agent to give up consumption smoothing and increase his consumption to the future.

Part 2.2: Labour Supply, Marshallian vs. Frisch Elasticities

We now apply the preference specification above to the households labour supply decision.

Suppose that at date t the household faces the within-period budget constraint

$$c_t = w_t n_t + I_t,$$

where w_t denotes the real wage and

$$I_t \equiv R_t a_t - a_{t+1}$$

is non-labour income net of savings.

4. Derive the intratemporal first-order condition characterizing optimal labour supply. Use this condition to obtain a labour supply rule of the form

$$n_t = n(c_t, w_t).$$

How does labour supply n_t respond to a change in the wage w_t when consumption c_t is held fixed? How does n_t respond to a change in consumption c_t when the wage w_t is held fixed?

Solution.

The problem of representative household is

$$\max_{(a_{t+1}, c_t, n_t)_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left(\frac{c_t^{1-\sigma} - 1}{1-\sigma} - \frac{n_t^{1+\phi}}{1+\phi}\right)$$
$$\text{s.t. } \forall t \in \mathbb{N}_0, c_t + a_{t+1} = w_t n_n + R_t a_t.$$

The Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left(\frac{c_t^{1-\sigma} - 1}{1-\sigma} - \frac{n_t^{1+\phi}}{1+\phi} + \lambda_t [w_t n_n + R_t a_t - c_t - a_{t+1}]\right).$$

The FOCs are

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \iff \beta^t c_t^{-\sigma} = \lambda_t \tag{2.1}$$

$$\frac{\partial \mathcal{L}}{\partial n_t} = 0 \iff \beta^t n_t^{\phi} = \lambda_t w_t \tag{2.2}$$

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}} = 0 \iff \lambda_{t+1} R_{t+1} = \lambda_t. \tag{2.3}$$

Equations (2.1) and (2.2) give the labor supply rule of the form

$$n_t = n_t(c_t, w_t) = \left(\frac{w_t}{c_t^{\sigma}}\right)^{\frac{1}{\phi}}. \tag{2.4}$$

Equations (2.4) is the labor supply rule. Since $c_t, \sigma, \phi > 0$, obviously we have that labor supply n_t increases in the wage w_t when consumption c_t is held fixed, and decreases in consumption c_t when the wage w_t is held fixed (the more you consume, the less you need to work). This condition pins down optimal labor supply. Working one extra hour raises income by w_t, which can be converted into consumption; the resulting utility gain is $u_c(c_t, n_t)\, w_t$. The utility cost is the marginal disutility of work, $-u_n(c_t, n_t)$. Optimality requires these two effects to balance, so that the marginal value of an extra hour worked equals its marginal cost in utility terms.

5. The *Marshallian elasticity* (or uncompensated elasticity) of labour supply with respect to the wage is defined as

$$\varepsilon_t^M \equiv \frac{d \log n_t}{d \log w_t},$$

where the total derivative allows c_t to adjust in response to w_t, holding I_t fixed.

This elasticity measures the total response of hours worked to a wage change, incorporating both substitution and income effects. Because the wage change is not offset by a compensating change in non-labour income, this elasticity is referred to as *uncompensated.*[1]

Using your labour supply rule, express ε_t^M in terms of $\frac{d \log c_t}{d \log w_t}$.

Next, holding I_t fixed, totally differentiate the budget constraint to show that

$$\frac{d \log c_t}{d \log w_t} = s_t^L + s_t^L \frac{d \log n_t}{d \log w_t},$$

where

$$s_t^L \equiv \frac{w_t n_t}{c_t} \in [0, 1]$$

denotes the labour income share in consumption. Provide an economic interpretation of each term on the right-hand side.

Combine these results to solve for the Marshallian elasticity

$$\varepsilon_t^M$$

as a function of σ, ϕ, and s_t^L. Carefully interpret the assumption that I_t is held fixed in light of the households optimization problem. Next, provide an intuitive discussion of

[1] By contrast, the *Hicksian* (or compensated) elasticity measures the response of hours worked to a wage change that is accompanied by an offsetting change in non-labour income, so as to keep utility constant. This compensation eliminates the income effect, so the Hicksian elasticity captures only the substitution effect: when the wage (the opportunity cost of leisure) increases, individuals substitute away from leisure toward work.

the signs of the substitution and income effects. Argue that ε_t^M may be either positive or negative, depending on which effect dominates (you are not required to separately identify the substitution and income effects).

Finally, consider the special case $s_t^L = 1$. Under what condition on σ do the substitution and income effects exactly offset each other?

Solution.

First, let's rewrite Equation (2.4) as follows:

$$c_t^{-\sigma} = \frac{n_t^\phi}{w_t}.$$

Now, take logs on both sides:

$$-\sigma \log c_t = \phi \log n_t - \log w_t,$$

and then:

$$\log n_t = \frac{1}{\phi} \log w_t - \frac{\sigma}{\phi} \log c_t.$$

Now, we can write:

$$\varepsilon_t^M = \frac{d \log n_t}{d \log w_t} = \frac{1}{\phi} - \frac{\sigma}{\phi} \frac{d \log c_t}{d \log w_t}.$$

Next, total differentiate the budget constraint keeping I_t fixed ($dI_t = 0$):

$$dc_t = n_t \; dw_t + w_t \; dn_t.$$

Now, notice that, given a variable x_t:

$$\frac{d \log x_t}{dx_t} = \frac{1}{x_t} \implies dx_t = x_t \; d \log x_t,$$

and so we can rewrite the differentiated budget constraint:

$$c_t \; d \log c_t = w_t n_t \; d \log w_t + w_t n_t \; d \log n_t.$$

Lastly, divide both sides by $c_t \; d \log w_t$:

$$\begin{aligned} \frac{d \log c_t}{d \log w_t} &= \frac{w_t n_t}{c_t} + \frac{w_t n_t}{c_t} \frac{d \log n_t}{d \log w_t} \\ &= s_t^L + s_t^L \frac{d \log n_t}{d \log w_t}. \end{aligned}$$

The first term, s_t^L, is the 'direct' or income effect of a higher wage on consumption holding hours fixed. The second term, $s_t^L \, d \log n_t / d \log w_t$, captures the additional change in consumption coming from the endogenous adjustment of hours in response to the wage change, that is, a substitution effect.

Therefore, we can rewrite the Marshallian elasticity:

$$\begin{aligned} \varepsilon_t^M = \frac{d \log n_t}{d \log w_t} &= \frac{1}{\phi} - \frac{\sigma}{\phi} \frac{d \log c_t}{d \log w_t} \\ &= \frac{1}{\phi} - \frac{\sigma}{\phi} \left(s_t^L + s_t^L \frac{d \log n_t}{d \log w_t} \right) \\ &= \frac{1}{\phi} - \frac{\sigma}{\phi} s_t^L - \frac{\sigma}{\phi} s_t^L \frac{d \log n_t}{d \log w_t}. \end{aligned}$$

Hence:

$$\left(1+\frac{\sigma}{\phi}s_t^L\right)\varepsilon_t^M = \frac{1}{\phi}-\frac{\sigma}{\phi}s_t^L \Rightarrow \varepsilon_t^M = \frac{1-\sigma s_t^L}{\phi+\sigma s_t^L}.$$

Here $I_t = R_t a_t - a_{t+1}$ is non-labor income net of saving. Holding I_t fixed means that we are holding fixed the household's asset position and saving choice in that period. So these effects that we are considering are static: when w_t changes, we have effects through the labor decision at time t, not through an endogenous adjustment of a_{t+1}, that will affect decisions in the next periods.

The sign of the Marshallian elasticity depends on which effect dominates:

- substitution effect: higher w_t raises the opportunity cost of leisure, so n_t tends to increase;
- income effect: higher w_t raises income, and agents want more leisure, so n_t tends to decrease.

In particular, the sign of ε_t^M depends only on the numerator:

- $\varepsilon_t^M > 0 \iff \sigma s_t^L < 1$
- $\varepsilon_t^M < 0 \iff \sigma s_t^L > 1$

If $s_t^L = 1$, then $\varepsilon_t^M = \frac{1-\sigma}{\phi+\sigma}$. Substitution and income effect exactly offset each other if $\sigma = 1$.

6. The Marshallian elasticity is fundamentally a static concept: it holds fixed the intertemporal allocation of resources. It is therefore particularly well suited for analyzing wage changes that are perceived as permanent. In your own words, briefly explain why a *permanent* wage change is the natural case for a static Marshallian elasticity.

   When wage changes are temporary and, importantly, induce adjustments in saving behavior, a different concept of elasticity is required that takes into account the shift in saving patterns. This motivates the *Frisch elasticity*. The Frisch elasticity measures the response of hours worked to a wage change while holding the marginal utility of wealth constant:

$$\varepsilon_t^F \equiv \left.\frac{\partial \log n_t}{\partial \log w_t}\right|_{\lambda_t}, \qquad \lambda_t \equiv u_c(c_t, n_t).$$

   It is the relevant concept to analyze how a transitory wage change affects hours. Small enough and temporary wage fluctuations have only a negligible effect on lifetime wealth. Thus it does not impacts the marginal utility of wealth. As a result, the Frisch elasticity is especially useful to assess the implications of wage fluctuations over the business cycle by isolating the intertemporal substitution response.

   Show that, for the utility function specified above, holding λ_t fixed is equivalent to holding c_t fixed. Compute ε_t^F explicitly as a function of ϕ.

   Solution.

   As we said in the previous point, Marshallian elasticity asks how much consumption c_t and labor supply n_t adjust when the wage w_t changes, holding the intertermporal decision fixed. A permanent change in the wage level doesn't create an incentive to move resources across time to smooth consumption: this is again the Permanent Income Hypothesis. If the wage is higher today and in the future, then lifetime income rises, an the household can adjust mainly through the intratemporal margin (work vs leisure today). By contrast, if the wage increase is temporary, the household would optimally use bonds to spread that

one-time income increase across many periods. Then, the Marshallian elasticity is no longer the right object.

Additionally, a permanent wage change is the natural case for a static Marshallian elasticity because the CRRA utility function is homothetic. If lifetime income increases or decreases, consumption increases or decreases proportionally, but the allocation of consumption (and hence investment) across time does not depend on the level of lifetime income (Krueger 2012).

The marginal utility of consumption is:

$$u_c(c_t, n_t) = \lambda_t = c_t^{-\sigma}$$

which depends only on c_t. Hence, holding fixed λ_t is equivalent to holding fixed c_t. Then, recall the Euler equation we derived in point 4:

$$c_t^{-\sigma} = \frac{n_t^{\phi}}{w_t}.$$

Taking logs:

$$-\sigma \log c_t = \phi \log n_t - \log w_t.$$

Now, total differentiate with respect to $\log w_t$ holding c_t fixed:

$$0 = \phi \frac{d \log n_t}{d \log w_t} - 1,$$

and so:

$$\frac{d \log n_t}{d \log w_t} = \varepsilon_t^F = \frac{1}{\phi}.$$

7. In the given setting, the intertemporal optimality condition, i.e., the Euler equation, is given by:

$$\frac{c_t^{-\sigma}}{c_{t+1}^{-\sigma}} = \beta R_{t+1}.$$

Combine this with your labour supply rule obtained in point 4. to obtain the intertemporal labour supply equation. Using the intertemporal labour supply equation, explain how expected wage growth (w_{t+1}/w_t) and the discounted return βR_{t+1} affect the evolution of hours (n_{t+1}/n_t).

Solution.

We know that $c_t^{-\sigma} = n_t^{\phi}/w_t$, so:

$$\frac{n_t^{\phi}}{n_{t+1}^{\phi}} \frac{w_{t+1}}{w_t} = \beta R_{t+1} \Rightarrow \frac{n_t^{\phi}}{n_{t+1}^{\phi}} = \beta R_{t+1} \frac{w_t}{w_{t+1}} \Rightarrow \left(\frac{n_{t+1}}{n_t}\right)^{\phi} = \frac{1}{\beta R_{t+1}} \frac{w_{t+1}}{w_t}$$

Therefore, higher expected wage growth increases labor hours growth. Intuitively, if wages are relatively higher tomorrow, households shift labor supply toward tomorrow (intertemporal substitution), so hours grow.

Higher discounted return βR_{t+1} decreases hours growth n_{t+1}/n_t. From the Euler equation for consumption typed in point 3., we know that higher βR_{t+1} implies higher c_{t+1}/c_t. This reduces the marginal benefit of working tomorrow, because consumption rises. Hence, n_{t+1}/n_t drops.

Exercise 3: Empirical Exercise

For this exercise, download the following series at monthly frequency from December 2000 through December 2025 from FRED:

- V = *Job Openings: Total Non-Farm* (Unit: Level in Thousands, Seasonally Adjusted);
  → `JTSJOL`
- U = *Unemployment Level* (Unit: Thousands of Persons, Seasonally Adjusted);
  → `UNEMPLOY`
- L = *Civilian labour Force Level* (Unit: Thousands of Persons, Monthly, Seasonally Adjusted);
  → `CLF16OV`
- U^s = *Number Unemployed for Less Than 5 Weeks* (Unit: Thousands of Persons, Seasonally Adjusted).
  → `UEMPLT5`

Part 3.1: Estimating Worker Flows

Consider the following dynamic equation for the level of unemployment:

$$U_{t+1} - U_t = p_t^{EU}(L_t - U_t) - p_t^{UE} U_t,$$

where p_t^{UE} is the job finding rate and p_t^{EU} a time-varying separation rate.

1. The aim of this exercise is to measure the inflow rate into unemployment (p_t^{EU}) and outflow rate from unemployment (p_t^{UE}) using the method proposed in "The Cyclical Behavior of Equilibrium Unemployment and Vacancies" by Shimer (2005). In your own words, summarize the approach detailed on pages 30-33, and briefly list advantages and limitations of the method.

   Solution.

   In his seminal 2005 paper, Robert Shimer develops a methodology to estimate the continuous-time transition rates between employment and unemployment, the job-finding rate f_t and the separation rate s_t, from discrete-time monthly CPS data.

   Core Identification Strategy. The key insight is that short-term unemployment (U_{t+1}^s, those unemployed for fewer than 5 weeks at date $t+1$) identifies workers who entered unemployment during month t. Workers remaining unemployed from month t to $t+1$ who were already unemployed at t are those who failed to exit unemployment. Under the assumption that all short-term unemployed at $t+1$ are new entrants during month t, the monthly probability that an unemployed worker fails to find a job is:

   $$F_t^c \equiv \frac{U_{t+1} - U_{t+1}^s}{U_t} \tag{3.1}$$

   so the discrete (monthly) job-finding probability is $F_t = 1 - F_t^c$. Assuming that, within each month, transitions follow a homogeneous Poisson process with constant hazard rates, the continuous-time job-finding rate is recovered via the mapping $f_t = -\ln(1 - F_t)$.

Recovery of the Separation Rate. To recover s_t, Shimer models the continuous-time evolution of unemployment within the month:

$$\dot{U}_{t+\tau} = s_t(L_t - U_{t+\tau}) - f_t\, U_{t+\tau}, \quad \tau \in [0, 1) \tag{3.2}$$

Holding L_t, f_t, and s_t constant during the month, this linear ODE has the closed-form solution:

$$U_{t+1} = \frac{s_t\, L_t}{f_t + s_t}\left(1 - e^{-(f_t+s_t)}\right) + e^{-(f_t+s_t)}\, U_t \tag{3.3}$$

Since this equation is implicit in s_t (given f_t, U_t, U_{t+1}, and L_t), it must be solved numerically, e.g. by root-finding.

Advantages:

(a) Relies on widely available aggregate CPS data rather than matched longitudinal records, which are susceptible to classification error and attrition.

(b) Explicitly corrects for time-aggregation bias, recovering hazard rates that more accurately reflect within-month transition dynamics.

Limitations

(a) Assumes constant transition rates within each month (piecewise-constant hazards).

(b) Abstracts from the participation margin: all flows between employment/unemployment and out-of-the-labour-force ($E \leftrightarrow N$, $U \leftrightarrow N$) are ignored, which can matter in recessions with discouraged-worker effects.

(c) Treats all short-term unemployed ($<$ 5 weeks) as new entrants, which slightly overstates inflows when some are re-entrants from prior months' short spells.

2. Next, calculate the probability of finding a job, p_t^{UE}, from December 2000 to December 2025, using data on the unemployment level and the short-term unemployment level, following Shimers method.

Solution.

Following Shimer's method, we calculate the monthly probability of finding a job, F_t (or p_t^{UE}), and the corresponding continuous-time job-finding rate f_t. We do this using data on the unemployment level (U_t) and short-term unemployment (U_t^s).

Table 3-1. Summary Statistics: Estimated Job Finding Rates (Dec 2000 – Dec 2025)

	Estimated Rates	
Statistic	**Job Finding Probability (F_t)**	**Job Finding Rate (f_t)**
Mean	0.435	0.582
Std. Dev.	0.082	0.124
Minimum	0.221	0.250
Maximum	0.655	0.985

Note: Estimates computed using the underlying methodology in Shimer (2005) correcting for time-aggregation bias.

3. Then, for the same period, compute the employment level using data on the labour force level and the unemployment level. With this, along with the short-term unemployment data, recover the separation rate, p_t^{EU}, by applying Shimers method. Explain why Shimer incorporates the job-finding rate p_t^{UE} in this calculation. Also, define and explain the concept of *time-aggregation bias*.

Solution.

Next, we compute the employment level E_t using data on the labour force L_t and the unemployment level U_t ($E_t = L_t - U_t$). With this information, along with the job finding rate f_t, we can recover the separation rate s_t (or p_t^{EU}). Shimer incorporates the job-finding rate f_t in the computation of s_t precisely because of time-aggregation bias, which requires accounting for workers who separate from employment *and* find new jobs within the same month. Without this correction, the separation rate would be underestimated.

```r
# Function to solve for separation rate s_t implicitly
solve_s <- function(U_t, U_t_plus_1, L_t, f_t) {
  # If any input is NA, return NA
  if (is.na(U_t) || is.na(U_t_plus_1) || is.na(L_t) || is.na(f_t)) return(
    NA_real_)

  # The objective function to find the root of (equation equals 0)
  obj_fun <- function(s) {
    U_t_plus_1 - ( (1 - exp(-f_t - s)) * s * L_t / (f_t + s) + exp(-f_t - s
    ) * U_t )
  }

  # Use uniroot to find s in a reasonable interval (e.g., 0.0001 to 0.5)
  res <- tryCatch(
    uniroot(obj_fun, interval = c(1e-6, 0.5))$root,
    error = function(e) NA_real_ # Return NA if no root is found
  )
  return(res)
}

# Apply the solver to each row
data <- data %>%
  mutate(
    # Compute the employment level based on labour force and unemployment
    E = L - U,
    U_lead = lead(U),
    # Calculate s_t row-wise
    s_t = pmap_dbl(list(U, U_lead, L, f_t), solve_s)
  ) %>%
  # Remove the temporary lead variable
  select(-U_lead)
```

Time-aggregation bias.

Definition 1 (Time-aggregation bias) Time-aggregation bias arises because discrete monthly snapshots of unemployment fail to register transitions that begin and end within the same observation period.

Example 1 For example, if a worker loses her job on the 5th of the month and finds a new one on the 20th, neither the separation nor the subsequent hire will appear in the stock data at the month's endpoints.

As a result, naive transition-rate estimates that compare consecutive monthly stocks systematically understate the true volume of both separations and job findings. Shimer's

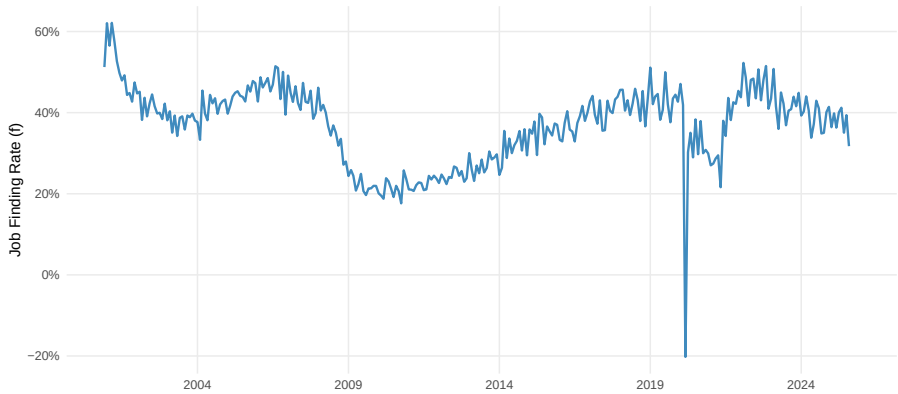
methodology corrects for this bias by modelling the within-month dynamics as a continuous-time Poisson process: (i) the monthly job-finding probability F_t is mapped to a continuous hazard rate $f_t = -\ln(1 - F_t)$, and (ii) the separation rate s_t is recovered by integrating the continuous-time unemployment differential equation over $[t, t+1]$, thereby accounting for the multiple transitions that may occur within a single discrete period.

Shimer incorporates the job-finding rate f_t in the computation of s_t precisely because time-aggregation bias requires accounting for workers who separate from employment and find new jobs within the same month. Without this correction, the separation rate would be underestimated.

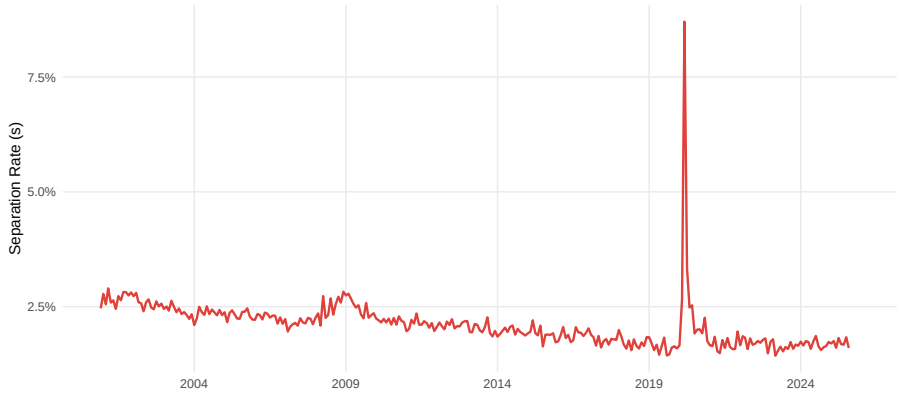
Because the equation for s_t is implicit, we can solve for it numerically month-by-month using a root-finding algorithm like `uniroot`.

4. Plot both time series for the job-finding rate and separation rate, and discuss their trends and cyclicality.

Solution.



(a) Job finding rate f



(b) Separation rate s

Figure 3-1. Monthly Transition Rates in the US labour Market. Continuous-time rates estimated from aggregate FRED data (Dec 2000 - Dec 2025). *Source: own calculations based on FRED data (UNEMPLOY, UEMPLT5, CLF16OV).*

Job-finding rate (f_t). The estimated job-finding rate is strongly **procyclical** and constitutes the dominant source of unemployment fluctuations over the business cycle. It declines sharply at the onset of each recession, falling by roughly 40-50% during the

Great Recession, and recovers only gradually during the subsequent expansion, consistent with the well-documented sluggishness of labour-market recoveries. The COVID-19 shock produced the sharpest single-month drop on record, followed by an exceptionally rapid recovery that pushed f_t to historically elevated levels by mid-2022, reflecting the extraordinary tightness of the post-pandemic labour market.

Separation rate (s_t). The separation rate is **countercyclical**, exhibiting brief but intense spikes at the onset of recessions. The spike associated with the Great Recession was large but brief; the COVID-19 spike was an order of magnitude larger yet even shorter-lived. Outside of recessionary episodes, s_t fluctuates within a narrow band, displaying comparatively little variation. This pattern confirms a central finding of Shimer (2005): while separation-rate spikes *initiate* rises in unemployment, it is the persistent decline in the job-finding rate that *sustains* elevated unemployment throughout the downturn.

Overall assessment. The time-series evidence supports a decomposition in which separations act as a "trigger" mechanism while the job-finding rate governs the depth and duration of unemployment cycles. This is consistent with the subsequent literature debate (e.g., Elsby, Michaels, and Solon (2009), who argue that separations play a somewhat larger role than Shimer (2005) initially suggested, particularly at the onset of downturns).

Part 3.2: The Beveridge Curve

In this exercise, we aim to analyze the Beveridge curve in the US, focusing on how it has evolved in recent years and the potential factors driving these changes.

1. Calculate the following labour market variables:

$$\text{Unemployment Rate}: \; u = \frac{U}{L}$$
$$\text{Employment Level}: \; E = L \cdot (1-u)$$
$$\text{Job Openings Rate}: \; v = \frac{V}{E+V}$$

Create a scatter plot with the unemployment rate (u) on the x-axis and the job vacancy rate (v) on the y-axis, for the following periods, using the specified colors:

- Dec 2000 - Dec 2007, (***Green***)
- Jan 2008 - June 2009, (***Orange***)
- July 2009 - Feb 2020, (***Red***)
- March 2020 - April 2022, (***Blue***)
- May 2022 - Dec 2025 (***Black***)

After plotting, discuss the similarities and, more importantly, the differences between these periods. Relate the movements along the Beveridge curve to the business cycles. Are there any significant shifts between these periods?

Solution.

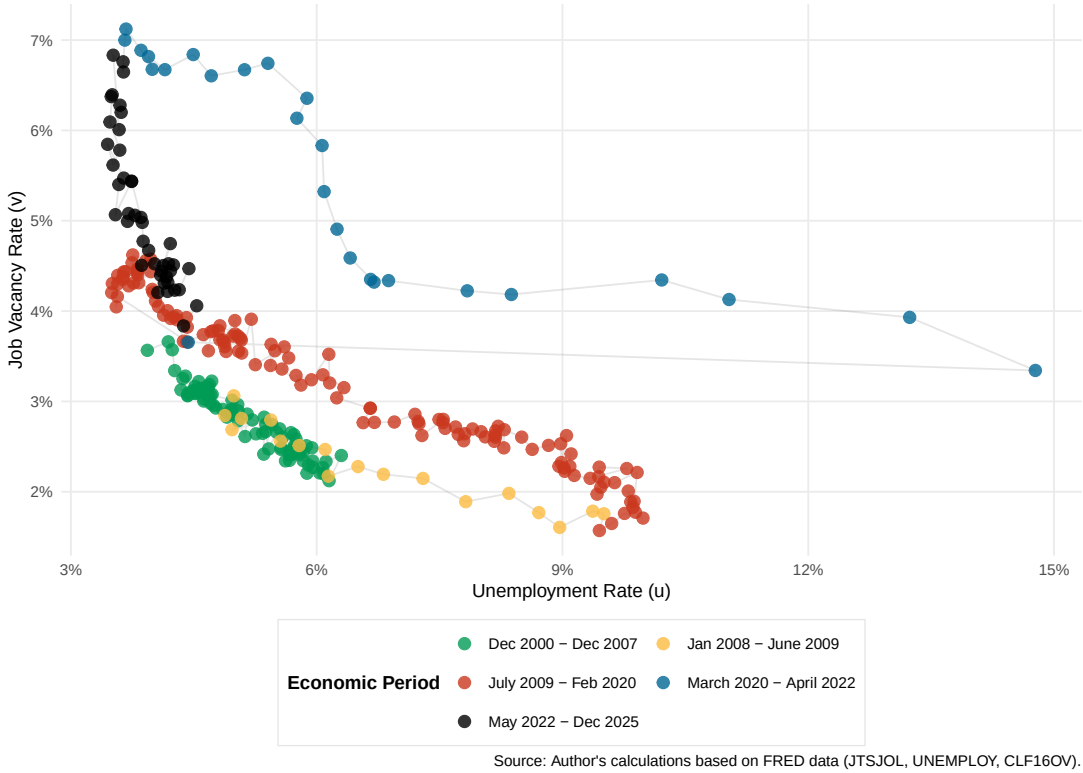


Figure 3-2. Dynamics of the US Beveridge Curve: Relationship between Unemployment Rate and Job Vacancy Rate

Movements along the curve. Within structurally stable regimes, e.g. the pre-crisis expansion (**green**, 2000-2007), the start of the Great Recession (**orange**, 2008-2009), and the post-crisis recovery (**red**, 2009-2020), the data trace out the canonical negative relationship predicted by the standard Beveridge curve. A key feature worth highlighting is that during the less turbulent period between 2000 and 2007 (in **green**), the curve displays a full cyclical movement: sliding down and to the right during the early 2000s contraction, and climbing back up and to the left along a roughly stable locus during the subsequent expansion.

Outward shifts and matching efficiency. A more striking feature is the outward shift of the curve between the pre-crisis (**green**) and post-crisis recovery (**red**) regimes. During the early recovery (2009-2013), the red cluster sits markedly to the right of the green locus at comparable vacancy rates, indicative of a deterioration in aggregate matching efficiency. This is broadly consistent with explanations involving skill mismatch, geographic mismatch, and extended unemployment benefit durations that reduced search intensity during that episode.

COVID-19 pandemic and recovery (blue). The pandemic produced an unprecedented pattern. The initial shock (March-April 2020) generated an extreme spike in unemployment with a simultaneous, sharp contraction in vacancies. However, the recovery trajectory is dramatically different from any prior cycle: vacancies surged to historic highs while unemployment rapidly declined, pushing the economy far to the upper-right of any previously observed Beveridge curve locus. This exceptional outward shift reflects a combination of sectoral reallocation shocks and the "Great Resignation", an unprecedented wave of voluntary job-to-job transitions that generated massive vacancy posting aimed at

replacing churning workers rather than expanding employment from the unemployment pool.

Monetary tightening and the "soft landing" (black). The most economically significant recent development is the trajectory of the black data points (May 2022 onward). As the Federal Reserve tightened monetary policy, the economy moved nearly vertically downward on the Beveridge curve: vacancies declined sharply from their historic peaks while the unemployment rate remained largely stable, showing only a slight outward drift at the very end of the series. This near-vertical descent is highly unusual and stands in stark contrast to the standard prediction that reducing labour-market tightness requires a severe trade-off between lower vacancies and higher unemployment. As Barlevy et al. (2024) argue, this "*immaculate disinflation*" is explained by the fact that the elevated vacancies reflected reallocation churn rather than fundamental labour demand: as the Great Resignation subsided, firms withdrew replacement postings without resorting to mass layoffs.

2. Read "The Shifting Reasons for Beveridge Curve Shifts" by Barlevy et al. (2024) with a focus on pages 94-97. How do the authors explain the shifts of the Beveridge curve during the Covid recession and recovery period (represented by the blue data points in your figure) and the unusual shape during the monetary tightening phase (represented by the black data points in your figure)? What is meant by the "Great Resignation"? How and why was the recovery from the pandemic recession different from past recoveries in terms of the Beveridge curve?

Solution.

Barlevy et al. (2024) decompose Beveridge-curve shifts into two distinct channels:

(a) Changes in **search efficiency**: the rate at which unemployed workers and vacant jobs form matches, holding the stock of each fixed.

(b) The intensity of **reallocation shocks**: the rate at which existing employment relationships dissolve, generating simultaneous inflows into both unemployment and vacancy posting.

Past recoveries. In the recoveries following the 2001 and 2007-09 recessions, Beveridge-curve shifts were primarily attributable to a decline in search efficiency. Long unemployment durations eroded human capital and weakened search effort, while geographic and sectoral mismatch between available workers and open positions reduced the matching rate. Extended unemployment insurance benefits may have further dampened search intensity. These factors caused unemployment to remain elevated even as vacancy posting recovered, producing the characteristic rightward shift of the curve.

The COVID-19 recovery and the "Great Resignation". The post-pandemic shift was driven by a fundamentally different mechanism. Barlevy et al. demonstrate that the dominant force was a surge in reallocation intensity, i.e. an unprecedented rise in voluntary employer-to-employer (job-to-job) transitions, popularly termed the "Great Resignation." As the economy reopened and labour demand surged, employed workers leveraged the tight market to pursue better wages, remote-work arrangements, and career changes. In a search-and-matching framework, this elevated churn requires firms to post a large number of vacancies merely to replace departing workers, even absent net employment growth. The vacancy-to-unemployment ratio therefore rose sharply, shifting the Beveridge curve

upward. Crucially, these vacancies were not aimed at drawing from the unemployment pool, but at competing for already-employed workers.

Monetary tightening and the near-vertical decline. The framework directly explains the unusual near-vertical trajectory of the black data points. The classical concern (Blanchard, Domash, and Summers 2022) was that restoring price stability would require a significant increase in unemployment, i.e. moving down a relatively flat Beveridge curve. Barlevy et al. show that this reasoning misidentifies the nature of the elevated vacancies. Because the pandemic-era vacancy surge was driven by reallocation churn rather than by fundamental excess demand for new hires, the tightening cycle merely unwound the churn: as quit rates normalized, firms withdrew replacement postings, and the vacancy rate fell rapidly. Since these vacancies had not been drawing workers from the unemployment pool in the first place, their disappearance did not generate layoffs or raise unemployment. The economy thus descended nearly vertically, achieving what commentators have termed an "immaculate disinflation" or "soft landing."

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